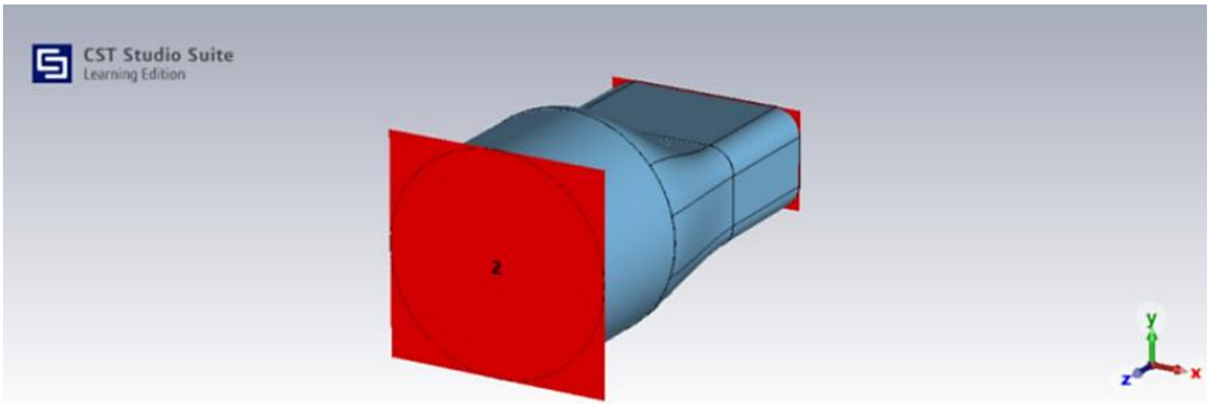
EXNO:13

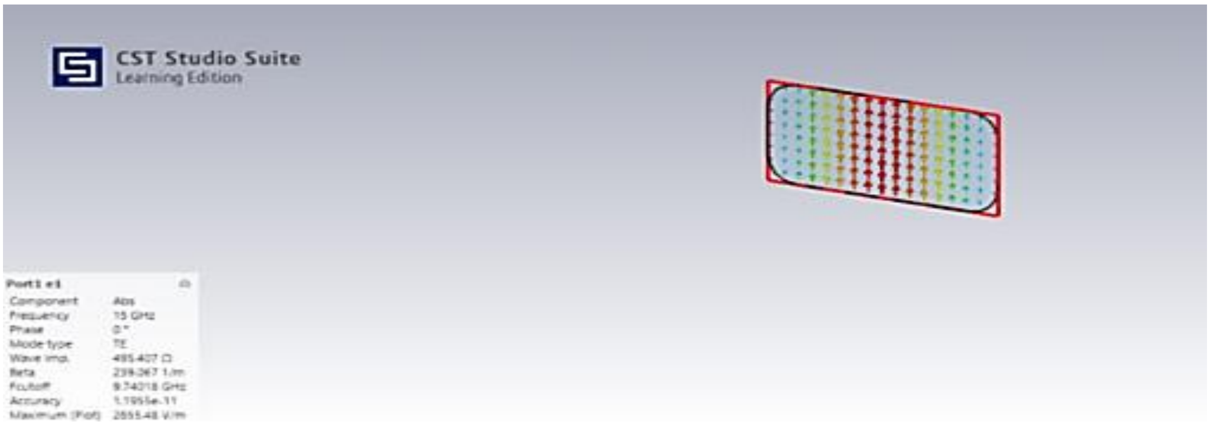
Investigation of electromagnetic field distribution in rectangular to circular waveguide transition.

Test Case 1:

Analyze the electric field and magnetic field distribution at Port 1 of the rectangular waveguide to circular waveguide transition and examine the electric and magnetic field parameters at Port 1 for studying electromagnetic field behavior and validating the system performance.

Simulation Results:





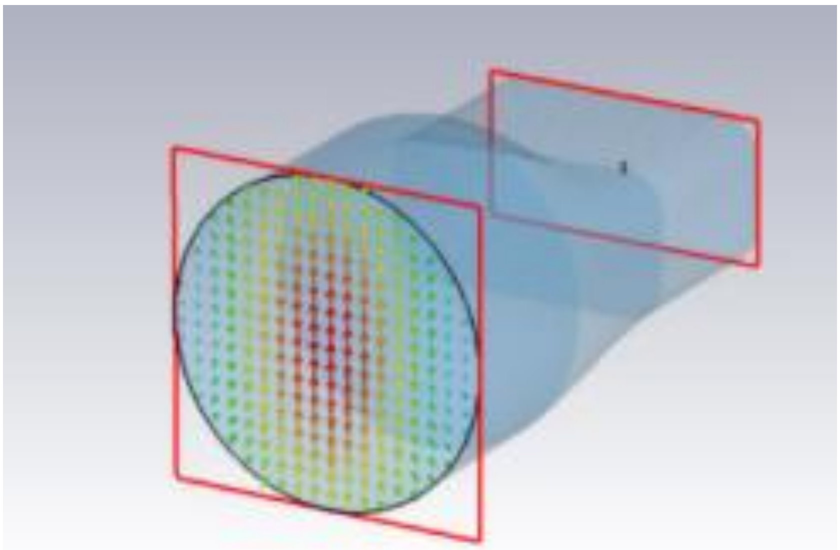
Output:

Parameter	Port 1
e1 (Electric Field)	TE mode
h1 (Magnetic Field)	TE mode
Frequency	15 GHz
Mode type	TE
Wave impedance	≈ 677 Ω
Phase constant β	≈ 213.5 1/m
Cutoff frequency	≈ 8.5 GHz
Peak Value	≈ 255.4 V/m

Test Case 2:

Analyze the electric field and magnetic field distribution at Port 2of the rectangular waveguide to circular waveguide transition and examine the electric and magnetic field parameters at Port 2 for studying electromagnetic field behavior and validating the system performance.

Simulation Result:



Output:

Parameter	**Port 2**
e2 (Electric Field)	**TE mode**
h2 (Magnetic Field)	**TE mode**
Frequency	**15 GHz**
Mode type	**TE**
Wave impedance	**≈ 537.3 Ω**
Phase constant β	**≈ 211.6 rad/m**
Cutoff frequency	**≈ 13.8 GHz**
Peak Value	**≈ 2.47 kV/m**